EFU WASTE MANAGEMENT LICENCE v1.0

Experimental EFU-based Waste Collection, Processing, and Disposal Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for waste management (collection, sorting, recycling, incineration, landfilling).

NOT a waste facility permit, hazardous waste authorization, or recycling quota replacement, but an open research protocol measuring waste operations' physical footprint in human-scale EFU metrics (energy, vehicles, landfill volume, leachate). Covers municipal, industrial, hazardous, and construction waste.

I. Core Concepts

1. **EFU-Waste (EFU-WASTE)**
   EFU framework application for waste management:
   - **EFU-E_WASTE**: diesel (collection trucks), electricity (sorting plants)
   - **EFU-M_WASTE**: sorting machinery wear, bags/containers
   - **EFU-V_WASTE**: landfill volume + incinerator ash
   - **EFU-L_WASTE**: leachate + emissions (dioxins from incineration)
   - **EFU-T_WASTE**: transport distance × waste volume
2. **Waste Sovereignty Gap (SS_WASTE)**
   SS_WASTE = TEFU_WASTE_total – Regional Waste Carrying Capacity (RWCC_WASTE)
   Measures if waste processing stays within regional landfill + incineration assimilation capacity.
3. **MROI_WASTE**
   MROI_WASTE = THI_WASTE / TEFU_WASTE
   THI_WASTE = recycling rate + avoided virgin materials + energy recovery.

II. EFU Waste Licence – Research Status

Documents EFU-audited waste companies, landfills, incinerators, recycling plants. **Does NOT replace** environmental permits, waste shipment rules, or EPR schemes. **Enables** incineration vs. landfill vs. recycling benchmarking.

III. Metrics and Thresholds

1. EFU Components (per tonne waste processed)

EFU-E_WASTE (Energy)
EFU-E_WASTE = energy_MWh / 12.88 / t_waste
Example: Collection = 0.1 MWh/t → 0.000008 EFU/t

EFU-V_WASTE (Volume)
EFU-V_WASTE = landfill_m³ / 100m³/person/year / t_waste
Example: Mixed MSW = 2m³/t → 0.00002 EFU/t

EFU-T_WASTE (Transport)
EFU-T_WASTE = truck_km / 10,000tkm/person/year / t_waste

TEFU_WASTE = 1.2×EFU-E + 1.5×EFU-M + 2.2×EFU-V + 1.8×EFU-L + 1.0×EFU-T

2. Regional Waste Carrying Capacity (RWCC_WASTE)

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RWCC_WASTE = Sustainable_t_waste / Population / 0.5t/person/year
Example: Germany = 50Mt / 84M / 0.5t = 1.19 EFU_WASTE/person/year
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IV. Licence Categories (EFU-WASTE Labels)

Level	SS_WASTE	MROI_WASTE	Label	Interpretation
1	≤ 0	≥1e-2	**EFU-WASTE SOVEREIGN**	Zero waste to landfill
2	0 < ≤25k EFU	≥5e-3	**EFU-WASTE BALANCED**	High recycling
3	25k < ≤125k EFU	≥1e-3	**EFU-WASTE DEFICIT**	Incineration dominant
4	>125k EFU	<1e-3	**EFU-WASTE CRITICAL**	Landfill dependent

V. Regional RWCC_WASTE Calibrations

Country	Waste (Mt/year)	Population (M)	RWCC_WASTE (EFU/person/year)
GERMANY	50	84	1.19
USA	300	340	0.44
CHINA	300	1,410	0.0007
JAPAN	40	125	0.26
EU	250	448	1.40

VI. Example: German Waste Management Sector

Input (annual sector):

- Total waste: 50Mt
- Landfilled: 1Mt (2%)
- Incinerated: 20Mt (40%)
- Recycled: 25Mt (50%)
- Collection diesel: 2B liters (7B MWh)
- Workers: 200,000 staff
- Region: Germany (RWCC_WASTE = 1.19 EFU/person/year)

Calculation:

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EFU-E_WASTE = 7B/12.88/50M = 0.011 EFU/t
EFU-V_WASTE = landfill vol/100/50M = 0.002 EFU/t

TEFU_WASTE_total ≈ 850,000 EFU/year
Allocated = 200k × 1.19 = 238,000 EFU/year
SS_WASTE = 850k – 238k = +612,000 EFU/year (Level 3: DEFICIT)

Result: EFU-WASTE DEFICIT
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Best Practice Comparison: Swedish model = 0.15 EFU/t (Level 1: SOVEREIGN)

VII. Implementation Notes

1. **EU Landfill Directive**: TEFU_WASTE for diversion target enforcement
2. **Extended Producer Responsibility**: EFU-M_WASTE basis for recycling fees
3. **Waste-to-energy tariffs**: EFU-E_WASTE metric for incineration incentives
4. **Data sources**: Tier 1 = weighbridge tickets; Tier 2 = Eurostat waste stats